G01 - AT1

Application Based Questions

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1) Given :-

Box	Red	Green	Blue	Total
A	6	4	2	12
B	5	7	3	15
C	8	2	5	15

Sol :-

Box B is twice as likely as A and C

$P(A) = 1/4$

$P(B) = 1/2$

$P(C) = 1/4$.

probability of blue ball

$P(Blue|A) = 1/3$

$P(Blue|B) = 7/15$

$P(Blue|C) = 2/15$.

using Baye's theorem.

$$P(C/Blue) = \frac{P(C)P(Blue|C)}{P(A)P(Blue|A) + P(B)P(Blue|B) + P(C)P(Blue|C)}$$

$$= \frac{\frac{1}{4} \times \frac{2}{15}}{\frac{1}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{7}{15} + \frac{1}{4} \times \frac{2}{15}}$$

$$= \frac{\frac{1}{30}}{\frac{1}{12} + \frac{7}{30} + \frac{1}{30}} = \frac{2}{21}$$

$P(C|Blue) = 9.52\%$.

2) Given:-

section	Red	Blue	Black	Total
A	5	3	2	10
B	4	6	5	15
C	7	2	6	15

Sol:-
section B is twice as likely as A and C

$P(A) = 1/4$
$P(B) = 1/2$
$P(C) = 1/4$

Probability of Blue
$P(Blue|A) = 3/10$
$P(Blue|B) = 6/15 = 2/5$
$P(Blue|C) = 2/15$

using Bayes theorem

$$P(C|Blue) = \frac{\frac{1}{4} \times \frac{2}{15}}{\frac{1}{4} \times \frac{3}{10} + \frac{1}{2} \times \frac{2}{5} + \frac{1}{4} \times \frac{2}{15}}$$

$$= \frac{\frac{1}{30}}{\frac{37}{120}} = \frac{4}{37}$$

$P(C|Blue) = 10.81\%$

3) Total testing samples
$1800 + 1800 = 3600$

20% testing $= 3600 \times 20\%$
$= 720$

$\therefore TP + TN + FP + FN = 720$
$210 + 190 + 30 + FN = 720$
$FN = 290$

Accuracy $= \frac{TP + TN}{Total} = \frac{210 + 190}{720}$

$= 55.56\%$

Precision $= \frac{TP}{TP + FP} = \frac{210}{210 + 30}$

$= 87.50\%$

Sensitivity $= \frac{TP}{TP + FN} = \frac{210}{210 + 290}$

$= 42.00\%$

Specificity $= \frac{TN}{TN + FP} = \frac{190}{190 + 30}$

$= 86.36\%$

F1 score $= 2 \times \frac{\text{Precision} \times \text{sensitivity}}{\text{Precision} + \text{sensitivity}}$

$= 2 \times \frac{0.875 \times 0.42}{0.875 + 0.42}$

$= 56.76\%$

4)

Epoch	Training loss	validation loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

9)

4) validation loss decreases until Epoch 3. From Epoch 4 training loss continues decreasing but validation loss increases.

∴ overfitting begins at Epoch 4.

ii) • Dropout - randomly disable some neuves during training
• L2 Regularization - penalizes large model weights and helps the model remain simpler.

iii) overfitting means the model learns the training data too closely, including loss and unnecessary patterns.
Therefore although training performance improves, performance on testdata decrease

5)

Epoch	Training Accuracy	validation Accuracy
1	60	58
2	65	63
3	70	68
4	75	72
5	80	73
6	85	72
7	88	70
8	92	68

a) i) Validation accuracy reaches its maximum of 73% at Epoch 5.

After Epoch 5,
- Training Accuracy increases
- Validation Accuracy decreases

∴ Overfitting starts at Epoch 6.

ii) The model is becoming too specialized to the training data.
Training accuracy increases, validation accuracy decreases
This indicates that model is learning training-specific patterns rather than general patterns.

iii) - Dropout:
- Early stopping.
Maybe L2 regularization, data augmentation

b)

Epoch	Training loss	Validation loss
1	3.0	2.7
2	2.6	2.3
3	2.3	2.1
4	2.1	2.0
5	1.9	2.2
6	1.7	2.5
7	1.6	2.8
8	1.4	3.1

6) i) Optimal generalization occurs when validation loss is minimum.
Minimum validation loss = 2.0
This occurs at
Epoch 4.

ii) After Epoch 4,
- Training loss continues decreasing.
- validation loss starts increasing.

This indicates overfitting.

iii) Dropout:
Randomly removes neurons during training, preventing the model.

L2 Regularization:
Add a penalty for large weights, encouraging a simpler model.

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7) Given:-
TP = 119 FP = 5
TN = 120 FN = 6

Sol:-

Accuracy = $\frac{119+120}{250}$ = 95.60%.

precision = $\frac{119}{119+5}$ = 95.97%.

sensitivity = $\frac{119}{119+6}$ = 95.20%.

Specificity = $\frac{120}{120+5}$ = 96.00%.

F1 score = $2 \times \frac{0.9597 \times 0.952}{0.9597+0.952}$ = 95.58%.

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1) Given:-

TP = 138 FN = 12

TN = 135 FP = 15

sol:-

Accuracy = $\frac{138+135}{300}$ = 91.00 %.

Precision = $\frac{138}{138+15}$ = 90.20 %.

sensitivity = $\frac{138}{138+12}$ = 92.00 %.

specificity = $\frac{135}{135+15}$ = 90.00 %.

F1 score = $\frac{2 \times 0.9020 \times 0.92}{0.9020 + 0.92}$ = 91.09 %.

2) Given:-

TP = 180 FN = 20

TN = 170 FP = 30

sol:-

Accuracy = $\frac{180+170}{400}$ = 87.50 %.

precision = $\frac{180}{180+30}$ = 85.71 %.

sensitivity = $\frac{180}{180+20}$ = 90.00 %.

specificity = $\frac{170}{170+30}$ = 85.00 %.

F1 score $= 2 \times \frac{0.8571 \times 0.90}{0.8571 + 0.90}$

$= 87.80\%$

10) Given:-

TP = 230 FN = 20

TN = 225 FP = 25

Sol:-

Accuracy $= \frac{230+225}{500} = 91.00\%$

Precision $= \frac{230}{230+25} = 90.20\%$

Sensitivity $= \frac{230}{230+20} = 92.00\%$

Specificity $= \frac{225}{225+25} = 90.00\%$

F1 score $= 2 \times \frac{0.9020 \times 0.92}{0.9020 + 0.92}$

$= 91.09\%$